

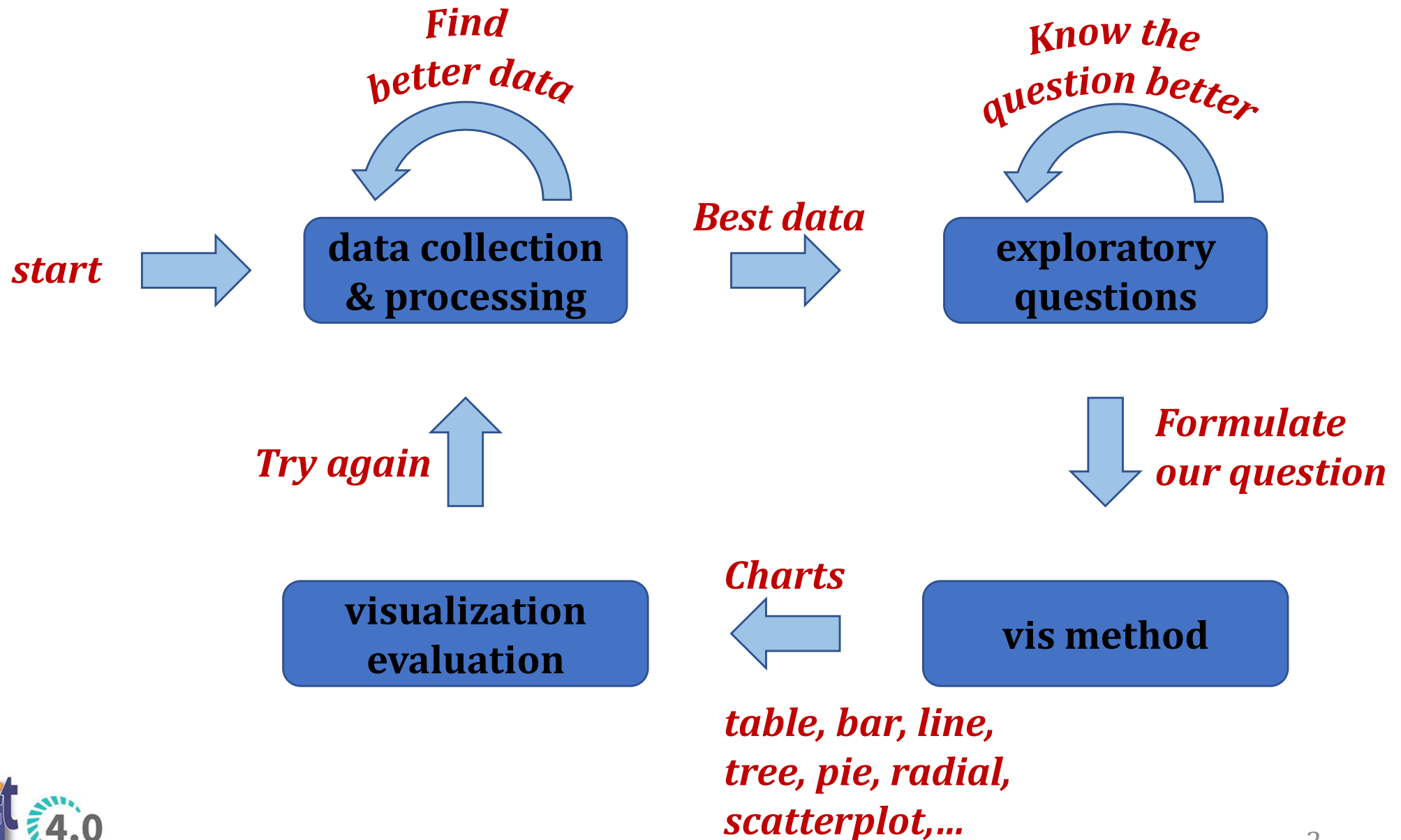
Data visualization

Chapter 03

Data abstraction



Visualization workflow



Data and dataset types

- The type of data is its *structural* or *mathematical interpretation*
 - EX: Integers or floats with +, *
 - ***Data level***
 - Item, link, attribute,...
 - ***Dataset level***
 - Data types are combined into a larger structure
 - Table, tree, field of values

Data types

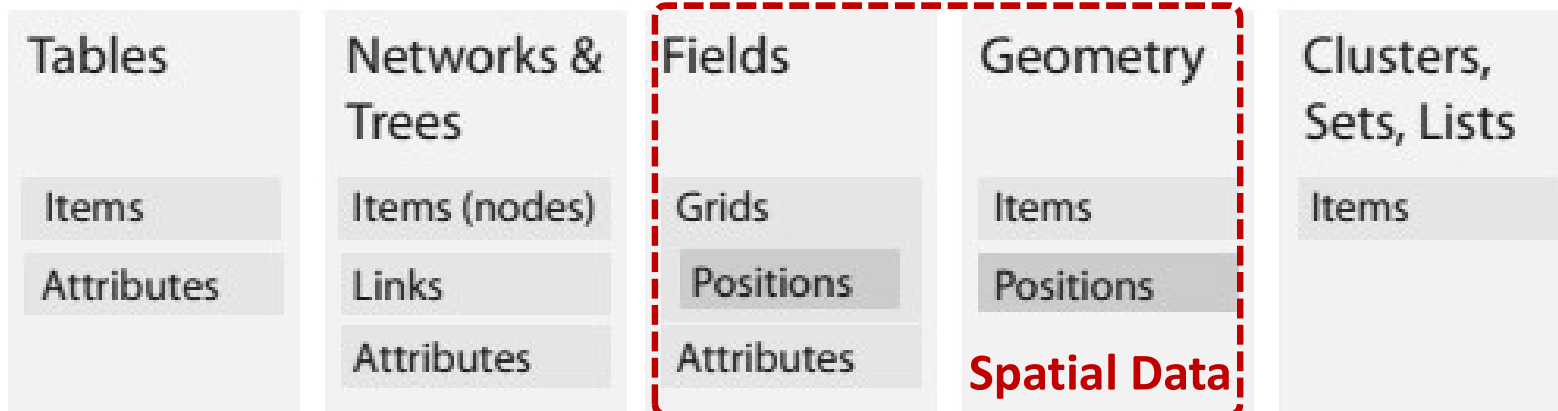


- There are five basic data types
 - *Attribute* – property that can be measured
 - Salary, price, temperature,...
 - *Item* – individual entity
 - A row of a table, a node in a network...
 - *Link* – relationship between items
 - *Position* – spatial/ location
 - Latitude–longitude,
 - *Grid* – how to samples continuous data
 - Geometric, topological

Dataset types

- A *dataset* is a collection of information that is the target of analysis
- There are four basic types
 - Tables
 - Networks
 - Fields
 - Geometry
- Complex combinations of multiple data types are commonly seen in real world applications.

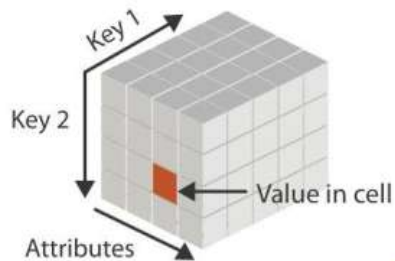
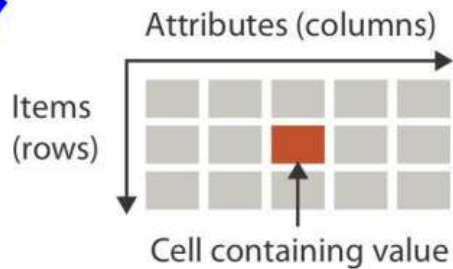
→ Data and Dataset Types



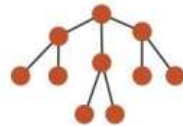
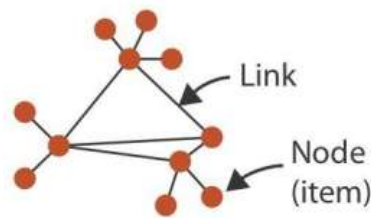
Dataset types

- There are four basic dataset types:

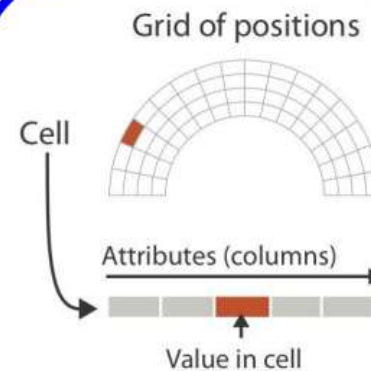
tables



networks



fields



geometry



Tamara Munzner, *Visualization Analysis and Design*, AK Peters, 2015

Items & Attributes

- **item**: individual entity, discrete
 - eg patient, car, stock, city
 - "*independent variable*"

Name	Age	Shirt Size	Favorite Fruit
Amy	8	S	Apple
Basil	7	S	Pear
Clara	9	M	Durian
Desmond	13	L	Elderberry
Ernest	12	L	Peach
Fanny	10	S	Lychee
George	9	M	Orange
Hector	8	L	Loquat
Ida	10	M	Pear
Amy	12	M	Orange

item: person

Items & Attributes

- **item**: individual entity, discrete
 - eg patient, car, stock, city
 - "*independent variable*"
- **attribute**: property that is measured, observed, logged...
 - eg height, blood pressure for patient
 - "*dependent variable*"

Name	Age	Shirt Size	Favorite Fruit
Amy	8	S	Apple
Basil	7	S	Pear
Clara	9	M	Durian
Desmond	13	L	Elderberry
Ernest	12	L	Peach
Fanny	10	S	Lychee
George	9	M	Orange
Hector	8	L	Loquat
Ida	10	M	Pear
Amy	12	M	Orange

item: person

Dataset types

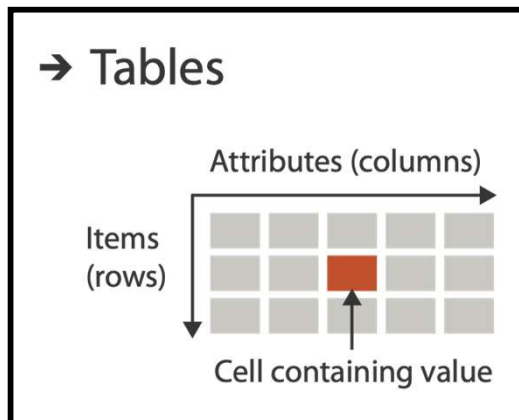
Tables

Items

Attributes

- **flat table**
 - one item per row
 - each column is attribute
 - cell holds value for item-attribute pair

attributes: name, age, shirt size, fave fruit



Name	Age	Shirt Size	Favorite Fruit
Amy	8	S	Apple
Basil	7	S	Pear
Clara	9	M	Durian
Desmond	13	L	Elderberry
Ernest	12	L	Peach
Fanny	10	S	Lychee
George	9	M	Orange
Hector	8	L	Loquat
Ida	10	M	Pear
Amy	12	M	Orange

item: person

Dataset types

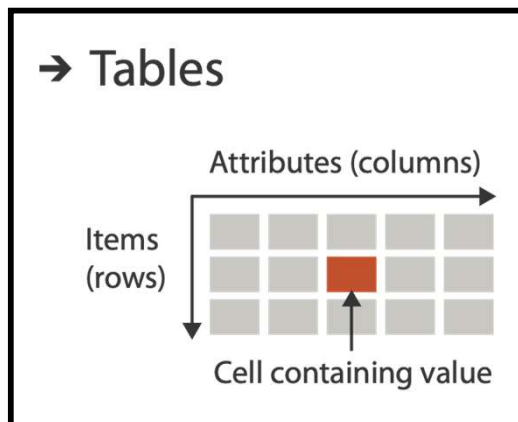
Tables

Items

Attributes

- **flat table**
 - one item per row
 - each column is attribute
 - cell holds value for item-attribute pair
 - unique key (*could be implicit*)

attributes: name, age, shirt size, fave fruit



ID	Name	Age	Shirt Size	Favorite Fruit
1	Amy	8	S	Apple
2	Basil	7	S	Pear
3	Clara	9	M	Durian
4	Desmond	13	L	Elderberry
5	Ernest	12	L	Peach
6	Fanny	10	S	Lychee
7	George	9	M	Orange
8	Hector	8	L	Loquat
9	Ida	10	M	Pear
10	Amy	12	M	Orange

item: person

Tables

A	B	C	S	T	U
Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
3	10/14/06	5-Low	Large Box	0.8	10/21/06
6	2/21/08	4-Not Specified	Small Pack	0.55	2/22/08
32	7/16/07	2-High	Small Pack	0.79	7/17/07
32	7/16/07	2-High	Jumbo Box	0.72	7/17/07
32	7/16/07	2-High	Medium Box	0.6	7/18/07
32	7/16/07	2-High	Medium Box	0.65	7/18/07
35	10/23/07	4-Not Specified	Wrap Bag	0.52	10/24/07
35	10/23/07	4-Not Specified	Small Box	0.58	10/25/07
36	11/3/07	1-Urgent	Small Box	0.55	11/3/07
65	3/18/07	1-Urgent	Small Pack	0.49	3/19/07
66	1/20/05	5-Low	Wrap Bag	0.56	1/20/05
69	6/4/05	4-Not Specified	Small Pack	0.44	6/6/05
69	6/4/05	4-Not Specified	Wrap Bag	0.6	6/6/05
70	12/18/06	5-Low	Small Box	0.59	12/23/06
70	12/18/06	5-Low	Wrap Bag	0.82	12/23/06
96	4/17/05	2-High	Small Box	0.55	4/19/05
97	1/29/06	3-Medium	Small Box	0.38	1/30/06
129	11/19/08	5-Low	Small Box	0.37	11/28/08
130	5/8/08	2-High	Small Box	0.37	5/9/08
130	5/8/08	2-High	Medium Box	0.38	5/10/08
130	5/8/08	2-High	Small Box	0.6	5/11/08
132	6/11/06	3-Medium	Medium Box	0.6	6/12/06
132	6/11/06	3-Medium	Jumbo Box	0.69	6/14/06
134	5/1/08	4-Not Specified	Large Box	0.82	5/3/08
135	10/21/07	4-Not Specified	Small Pack	0.64	10/23/07
166	9/12/07	2-High	Small Box	0.55	9/14/07
193	8/8/06	1-Urgent	Medium Box	0.57	8/10/06
194	4/5/08	3-Medium	Wrap Bag	0.42	4/7/08

item

Tables

A	B	C	S	T	U
Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
3	10/14/06	5-Low	Large Box	0.8	10/21/06
6	2/21/08	4-Not Specified	Small Pack	0.55	2/22/08
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32	7/16/07	2-High	Jumbo Box	0.72	7/17/07
32	7/16/07	2-High	Medium Box	0.6	7/18/07
32	7/16/07	2-High	Medium Box	0.65	7/18/07
35	10/23/07	4-Not Specified	Wrap Bag	0.52	10/24/07
35	10/23/07	4-Not Specified	Small Box	0.58	10/25/07
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69	6/4/05	4-Not Specified	Wrap Bag	0.6	6/6/05
70	12/18/06	5-Low	Small Box	0.59	12/23/06
70	12/18/06	5-Low	Wrap Bag	0.82	12/23/06
96	4/17/05	2-High	Small Box	0.55	4/19/05
97	1/29/06	3-Medium	Small Box	0.38	1/30/06
129	11/19/08	5-Low	Small Box	0.37	11/28/08
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134	5/1/08	4-Not Specified	Large Box	0.82	5/3/08
135	10/21/07	4-Not Specified	Small Pack	0.64	10/23/07
166	9/12/07	2-High	Small Box	0.55	9/14/07
193	8/8/06	1-Urgent	Medium Box	0.57	8/10/06
194	4/5/08	3-Medium	Wrap Bag	0.42	4/7/08

item

attribute

Tables

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3	10/14/06	5-Low	Large Box	0.8	10/21/06
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32	7/16/07	2-High	Medium Box	0.6	7/18/07
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35	10/23/07	4-Not Specified	Small Box	0.58	10/25/07
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135	10/21/07	4-Not Specified	Small Pack	0.64	10/23/07
166	9/12/07	2-High	Small Box	0.55	9/14/07
193	8/8/06	1-Urgent	Medium Box	0.57	8/10/06
194	4/5/08	3-Medium	Wrap Bag	0.42	4/7/08

item

cell

attribute

Tables

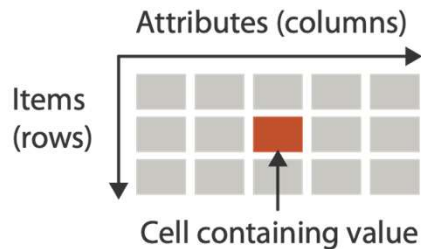
Tables

Items

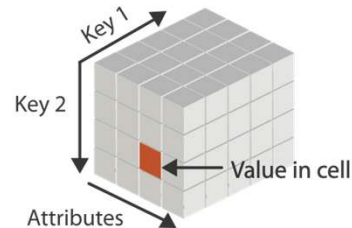
Attributes

- **multidimensional tables**
 - indexing based on multiple keys
 - eg genes, patients

→ Tables



→ Multidimensional Table



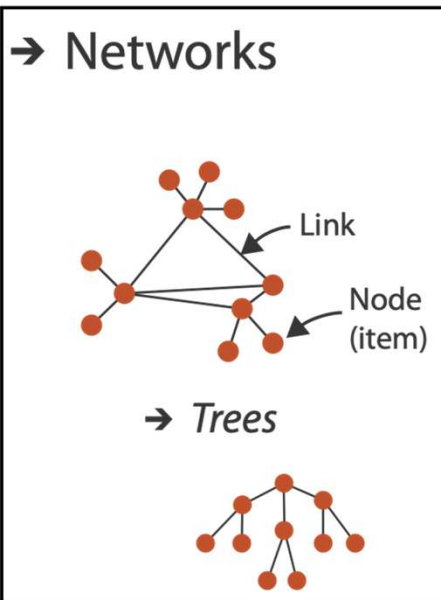
	A	B	C	D	E
1	A	B	C	D	E
2	1	A	B	C	E
3	2	A	B	C	E
4	3	A	B	C	E
5	4	A	B	C	E
6	5	A	B	C	E
7	6	A	B	C	E
8	7	A	B	C	E
9	8	A	B	C	E
10	9	A	B	C	E
11	10	A	B	C	E
12	11	A	B	C	E
13	12	A	B	C	E
14	13	A	B	C	E
15	14	A	B	C	E
16	15	A	B	C	E
17	16	A	B	C	E
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37	36	A	B	C	E
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281	280	A	B	C	E
282	281	A	B	C	E
283	282	A	B	C	E
284	283	A	B	C	E
285	284	A	B	C	E
286	285	A	B	C	E
287	286	A	B	C	E
288	287	A	B	C	E
289	288	A	B	C	E
290	289	A	B	C	E
291	290	A	B	C	E
292	291	A	B	C	E
293	292	A	B	C	E
294	293	A	B	C	E
295	294	A	B	C	E
296	295	A	B	C	E
297	296	A	B	C	E
298	297	A	B	C	E
299	298	A	B	C	E
300	299	A	B	C	E
301	300	A	B	C	E
302	301	A	B	C	E
303	302	A	B	C	E
304	303	A	B	C	E
305	304	A	B	C	E
306	305	A	B	C	E
307	306	A	B	C	E
308	307	A	B	C	E
309	308	A	B	C	E
310	309	A	B	C	E
311	310				

Tables

- The 3D table uses three indices to find a cell

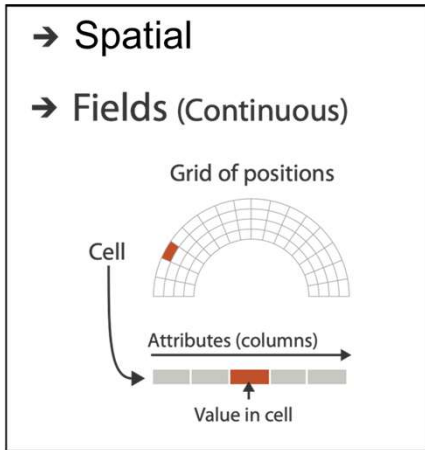
Department		A	B	C	D	E	F
Male	Accepted	512	353	120	138	53	22
	Rejected	313	207	205	279	138	351
Female	Accepted	89	17	202	131	94	24
	Rejected	19	8	391	244	299	317

Dataset types - Networks



- **Networks** (or graphs) are useful to represent *relationship between several items*.
 - Item is a *node*
 - Each node can have associated *attributes* (e.g., city size)
 - *Link* is a relation between two items.
 - Each link may also have associated *attributes* (e.g., distance between two cities).
- A **tree** is just a special type of networks.

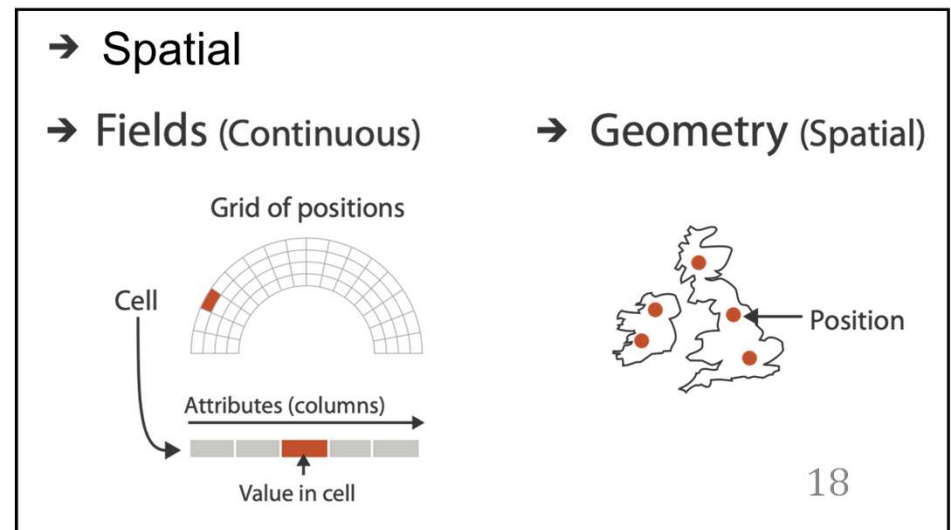
Dataset types - Fields



- A *field* contains attribute values associated with cells.
 - Each *cell* contains measurements or calculations from a continuous domain.
 - Obtaining values from a *continuous* domain is usually very challenging because the domain is a continuum.
- A good *sampling* strategy for taking measurements from discrete positions is needed.

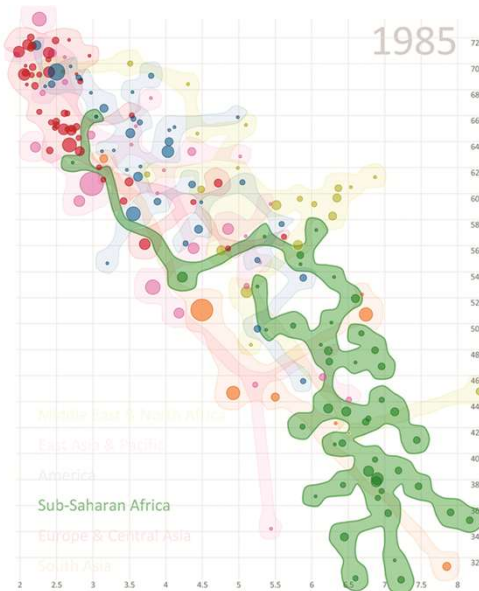
Dataset types - Geometry

- The *geometry* dataset type specifies information about the shape of items with explicit spatial information.
 - The *items* could be points, lines/curves, 2D surfaces/regions, 3D volumes, or even higher dimensional data.
 - Therefore, geometry datasets are spatial, and typically occur in the context of tasks that require shape understanding.

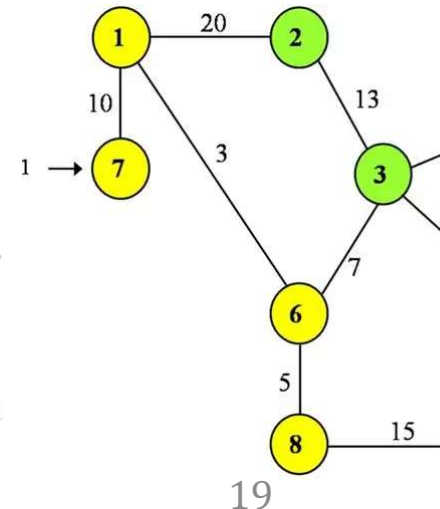
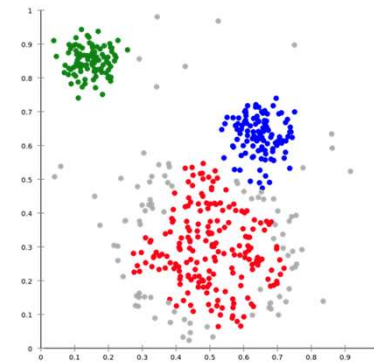


Other multiple combinations

- Set – unordered group
- List – ordered set
- Cluster – group based on similarity of some attribute
- Path – ordered set of segments in a network or tree



Rank	School Name	Academic repu
	Filter: <None>	
1.	Massachusetts Inst	
2.	University of Camb	
3.	Harvard University	100 (1)
4.	UCL (University Co	
5.	University of Oxfor	
6.	Imperial College L	
7.	Yale University	
8.	University of Chic	



Dataset availability



- There are two kinds of dataset availability: *static* and *dynamic*
- *Static file* means the entire dataset is available all at once
- *Dynamic streams* mean a dataset is continuously generated, by different sources, without having access to all the data

Attribute types: categorical vs ordered

- **Categorical (*nominal*)**

- Equal or not equal to other values
- Ex: gender, student number, course name,...
- No concept of relative relation other than inclusion in the set

- **Ordered**

- Implicit ordering
- There are two ordered types: ordinal and quantitative
 - **Ordinal** – well-defined ordering but without full-fledged arithmetic
 - **Quantitative** – well-defined capability to perform arithmetic and comparison

→ Categorical



e.g.,
fruit (apple, pear, grape),
colleges (CAMD, CCIS, COE)

→ Ordered

→ *Ordinal*



e.g.,
sizes (xs, s, m, l, xl),
months (J, F, M)

→ *Quantitative (continuous)*



e.g.,
lengths (1', 2.5', 5'),
population

Ordered attribute type: **direction**

- An ordered dataset can be *sequential* or *diverging*.
 - A *sequential* dataset has a homogeneous range from a minimum to a maximum.
 - A *diverging* dataset has data measured from a based point and extends to both ends
 - A *cyclic* dataset has its values wrapped around back to a starting point

→ Sequential



→ Diverging



→ Cyclic



Attributes: hierarchical

- An attribute may have a hierarchical structure.
- Ex:
 - Date sold products: 15/10/2022 (*weeks, months, years*)
 - A geographic attribute of a postal code may be aggregated to the level of cities, states,...

Cần Thơ		Ninh Kiều		An Hòa	
9	0	1	0	5	0
Province/ City		district/ town		ward	

Attribute types: categorical vs ordered



- Continuous
 - Data can take any value within the range
 - Ex: Number grade
- Discrete
 - Data can take only certain values
 - Ex: Letter grade (A+)
- We can convert between these using *clustering* or *thresholds*

Attribute types: categorical vs ordered

A	B	C	S	T	U
Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
3	10/14/06	5-Low	Large Box	0.8	10/21/06
6	2/21/08	4-Not Specified	Small Pack	0.55	2/22/08
32	7/16/07	2-High	Small Pack	0.79	7/17/07
32	7/16/07	2-High	Jumbo Box	0.72	7/17/07
32	7/16/07	2-High	Medium Box	0.6	7/18/07
32	7/16/07	2-High	Medium Box	0.65	7/18/07
35	10/23/07	4-Not Specified	Wrap Bag	0.52	10/24/07
35	10/23/07	4-Not Specified	Small Box	0.58	10/25/07
36	11/3/07	1-Urgent	Small Box	0.55	11/3/07
65	3/18/07	1-Urgent	Small Pack	0.49	3/19/07
66	1/20/05	5-Low	Wrap Bag	0.56	1/20/05
69	6/4/05	4-Not Specified	Small Pack	0.44	6/6/05
69	6/4/05	4-Not Specified	Small Pack	0.6	6/6/05
70	12/18/06	5-Low		0.59	12/23/06
70	12/18/06	5-Low		0.82	12/23/06
96	4/17/05	2-High		0.55	4/19/05
97	1/29/06	3-Medium		0.38	1/30/06
129	11/19/08	5-Low		0.37	11/28/08
130	5/8/08	2-High	Small Box	0.37	5/9/08
130	5/8/08	2-High	Medium Box	0.38	5/10/08
130	5/8/08	2-High	Small Box	0.6	5/11/08
132	6/11/06	3-Medium	Medium Box	0.6	6/12/06
132	6/11/06	3-Medium	Jumbo Box	0.69	6/14/06
134	5/1/08	4-Not Specified	Large Box	0.82	5/3/08
135	10/21/07	4-Not Specified	Small Pack	0.64	10/23/07
166	9/12/07	2-High	Small Box	0.55	9/14/07
193	8/8/06	1-Urgent	Medium Box	0.57	8/10/06
194	4/5/08	3-Medium	Wrap Bag	0.42	4/7/08

quantitative
ordinal
categorical

Data **semantics**



- **Problem**

- Raw data is meaningless without context.

- **Example:**

14, 2.6, 30, 30, 15, 100001

- Are these 3D spatial coordinates?
- Two 2D points connected by a weighted link?

- **Conclusion**

- Numbers alone tell us nothing
⇒ We need **Semantics** attached.

Data semantics

- **Definition**

Semantics = Real-world meaning attached to data (Metadata).

- **The "Basil 7 S Pear" Puzzle:**

- *Data:* string "Basil 7 S Pear".
- *Hypothesis:*
 - Sales: A food shipment (Basil + Pears) arriving on the 7th?
 - Services: "Basil Point" neighborhood has 7 inches of snow, cleared by "Pear Creek" company?
 - Experiment: A lab rat named "Basil" took 7 tries to run the "South" maze for a "Pear" reward?

⇒ Visualization requires understanding semantics to choose the right display.

Data semantics

- semantics: real-world meaning

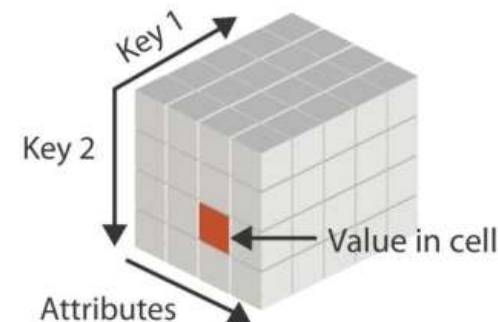
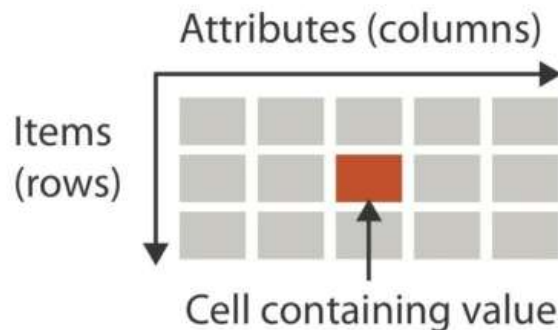
Name	Age	Shirt Size	Favorite Fruit
Amy	8	S	Apple
Basil	7	S	Pear
Clara	9	M	Durian
Desmond	13	L	Elderberry
Ernest	12	L	Peach
Fanny	10	S	Lychee
George	9	M	Orange
Hector	8	L	Loquat
Ida	10	M	Pear
Amy	12	M	Orange

Data semantics – Key and Value

- **Key** semantics

- A **key** attribute is an index that is used to find **value** attributes
- In a table, the key may just be *explicit* or *implicit* (indices)
- Keys may be *categorical* or *ordinal* attributes (not *quantitative*)

- **Value** attribute is a value of the table cell



Data semantics – Key and Value

- Flat tables

A	B	C	S	T	U
Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
3	10/14/06	5-Low	Large Box	0.8	10/21/06
6	2/21/08	4-Not Specified	Small Pack	0.55	2/22/08
32	7/16/07	2-High	Small Pack	0.79	7/17/07
32	7/16/07	2-High	Jumbo Box	0.72	7/17/07
32	7/16/07	2-High	Medium Box	0.6	7/18/07
32	7/16/07	2-High	Medium Box	0.65	7/18/07
35	10/23/07	4-Not Specified	Wrap Bag	0.52	10/24/07
35	10/23/07	4-Not Specified	Small Box	0.58	10/25/07
36	11/3/07	1-Urgent	Small Box	0.55	11/3/07
65	3/18/07	1-Urgent	Small Pack	0.49	3/19/07
66	1/20/05	5-Low	Wrap Bag	0.56	1/20/05
69	6/4/05	4-Not Specified	Small Pack	0.44	6/6/05
69	6/4/05	4-Not Specified		0.6	6/6/05
70	12/18/06	5-Low		0.59	12/23/06
70	12/18/06	5-Low		0.82	12/23/06
96	4/17/05	2-High		0.55	4/19/05
97	1/29/06	3-Medium		0.38	1/30/06
129	11/19/08	5-Low		0.37	11/28/08
130	5/8/08	2-High	Small Box	0.37	5/9/08
130	5/8/08	2-High	Medium Box	0.38	5/10/08
130	5/8/08	2-High	Small Box	0.6	5/11/08
132	6/11/06	3-Medium	Medium Box	0.6	6/12/06
132	6/11/06	3-Medium	Jumbo Box	0.69	6/14/06
134	5/1/08	4-Not Specified	Large Box	0.82	5/3/08
135	10/21/07	4-Not Specified	Small Pack	0.64	10/23/07
166	9/12/07	2-High	Small Box	0.55	9/14/07
193	8/8/06	1-Urgent	Medium Box	0.57	8/10/06
194	4/5/08	3-Medium	Wrap Bag	0.42	4/7/08

quantitative
ordinal
categorical

Data semantics – Key and Value

- Multidimensional tables
 - Multiple keys are required to look up an item
 - The combination of all keys must be unique for each item

Department		A	B	C	D	E	F
Male	Accepted	512	353	120	138	53	22
	Rejected	313	207	205	279	138	351
Female	Accepted	89	17	202	131	94	24
	Rejected	19	8	391	244	299	317

Data abstraction

- Data source

1	Tỉnh	Tổng số ca lây nhiễm	Số ca lây nhiễm mới	Tổng số ca tử vong	Ngày	Nhóm tuổi
2	Hồ Chí Minh	112215	2563	2105	8/6/2021	50 - 80
3	Bình Dương	22689	322	144	8/6/2021	30 - 40
4	Long An	8451	286	84	8/6/2021	1 - 18
5	Đồng Nai	6347	207	39	8/6/2021	18 - 30
6	Bắc Giang	5820	0	13	8/6/2021	50 - 80
7	Khánh Hòa	3214	53	81	8/6/2021	30 - 40
8	Đồng Tháp	2.96	0	24	8/6/2021	1 - 18
9	Tiền Giang	2770	253	75	8/6/2021	18 - 30
64	Hồ Chí Minh	115548	1836	2324	8/7/2021	30 - 40
65	Bình Dương	24418	882	145	8/7/2021	1 - 18
66	Long An	9184	160	104	8/7/2021	18 - 30
67	Đồng Nai	7160	466	53	8/7/2021	30 - 40
68	Bắc Giang	5820	0	13	8/7/2021	1 - 18
69	Khánh Hòa	3313	11	81	8/7/2021	18 - 30
70	Đồng Tháp	3229	0	27	8/7/2021	50 - 80
71	Tiền Giang	2935	165	79	8/7/2021	30 - 40
72	Tây Ninh	2479	0	6	8/7/2021	1 - 18

Data abstraction



- **Data type:** item/attribute/position/link/grid
- **Dataset type:** table/ field/network/geometry
- **Dataset availability:** static / dynamic
- **Item semantic:** meaning of item

Attribute Name	Type (C,O,Q)	Key/value	Direction	Hierarchical	Continuous/ Discrete	Semantic



Q&A



Chapter 03

Task Abstraction

